QX4001 – Molecules to Materials

Rotation 2 – Question Set 2

1) Identify which of the following species are acids, and what acid-base definitions they comply with: H_2SO_4, HCl, NaH, CH_3COOH, CH_4, $AlCl_3$. Write Lewis structures for these compounds.

- Arrhenius acid:
  - An acid is a substance that, when dissolved in water, increases the concentration of H^+ ions.
- Brønsted-Lowry acid:
  - A substance that donates H^+ to another substance (base).
- Lewis acid:
  - A substance that is an electron pair acceptor

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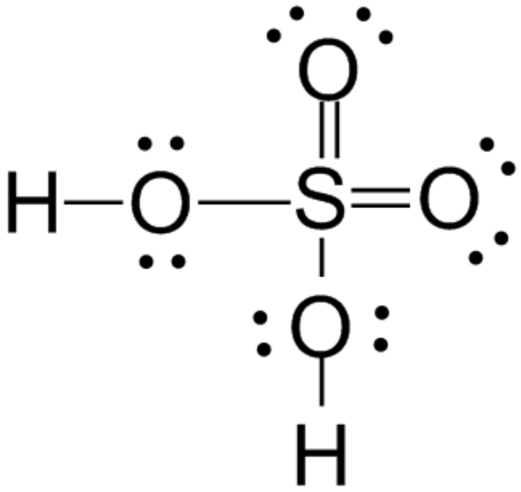
- H_2SO_4
  - Arrhenius acid
  - Brønsted-Lowry acid
  - Lewis Acid



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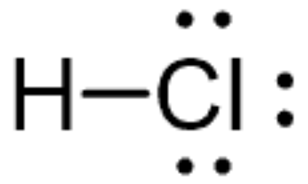
- HCl
  - Arrhenius acid
  - Brønsted-Lowry acid
  - Lewis Acid



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- NaH
  - Arrhenius Base
  - Brønsted-Lowry Base
  - Lewis Base

$[Na]^+ [:H]^-$

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- CH_3COOH
  - Arrhenius acid
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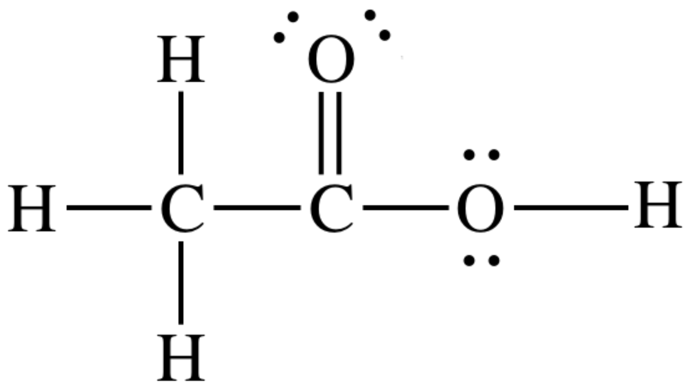

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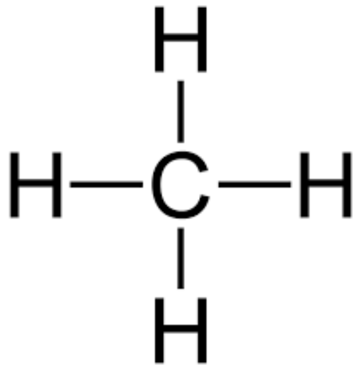
- CH_4
  - Not an acid or base



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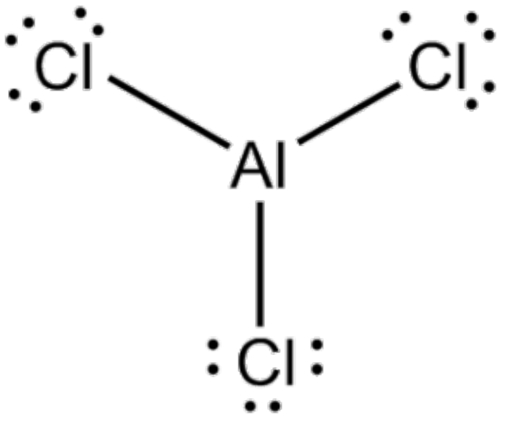
- $AlCl_3$
  - Lewis Acid



2) Given the broad Lewis Acid-Base definitions, can all reactions be classified as acid-base reactions?

- Not all reactions involve electron pair donors or acceptors
  - Radical Reactions
  - Redox reactions

3) Calculate the pH of the following aqueous solutions:

a) 0.3M, 0.0078M and 2.3M solutions of HCl in water (*pKa*=-7) (compare the values with full dissociation)

b) 0.3M solution of H_2SO_4 (K_{a1} is extremely large, $K_{a2} = 1.2 \times 10^{-2}$)

c) Compare the value calculated in b with an assumption that both Ka values for H_2SO_4 is large.

d) 0.4M solution of NaOH

3) Calculate the pH of the following aqueous solutions:

a) 0.3M, 0.0078M and 2.3M solutions of HCl in water (*pKa*=-7) (compare the values with full dissociation)

- $[H^+] = 0.3\text{M}$ So $pH = -log[H^+] = 0.52$
- $[H^+] = 0.0078\text{M}$ So $pH = -log[H^+] = 2.10$
- $[H^+] = 2.3\text{M}$ So $pH = -log[H^+] = -0.36$

3) Calculate the pH of the following aqueous solutions:

b) 0.3M solution of H_2SO_4 (K_{a1} is extremely large, K_{a2} = 1.2 x 10^{-2})

$$K_{a1} = \frac{[H^+][HSO_4^-]}{[H_2SO_4]} = very\ large \qquad initial\ concentration\ of\ 0.3M$$

$$K_{a2} = \frac{[H^+][SO_4^{2-}]}{[HSO_4^-]} = 1.2\times10^{-2}M = \frac{[0.3+x][x]}{[0.3-x]}\ this\ gives\ us\ x^2 + 0.312x - 0.0036 = 0$$

$$x_1 = -0.323\ or\ x_2 = 0.011$$

So we find that $[H^+] = 0.3M + 0.011M = 0.311M\ so\ pH = -log[H^+] = 0.507$

3) Calculate the pH of the following aqueous solutions:

c) Compare the value calculated in b with an assumption that both Ka values for H_2SO_4 is large.

Assuming that both $K_a's$ are very large we find that $[H^+] = 0.6M$

$$pH = -log[H^+] = -log[0.6]$$
$$= 0.222$$

3) Calculate the pH of the following aqueous solutions:

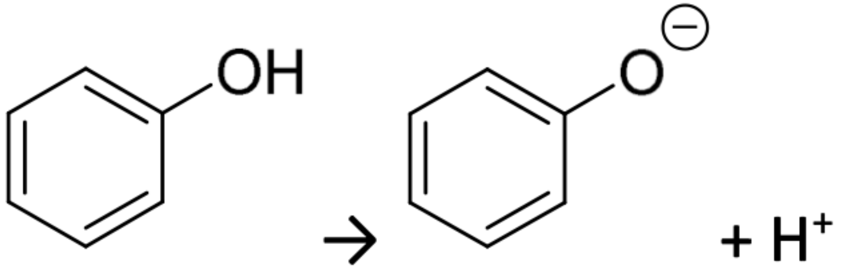
d) 0.4M solution of NaOH

- $[OH^-] = 0.4\text{M}$ So $pOH = -log[OH^-] = 0.398$
- $pH = 14 - pOH = 14 - 0.398 = 13.6$
- Or $K_c = [H^+][OH^-]$ so $[H^+] = {}^{K_c}/_{[OH^-]} = \frac{10^{-14}}{0.4}$
- $[H^+] = 2.5\times10^{-14}$ $so\ pH = -log\ [H^+] = 13.6$

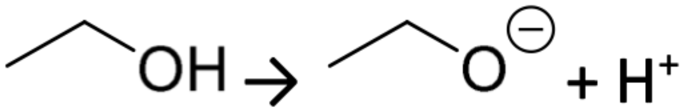
4) Explain the differences below using resonant forms:

i. The reaction $H_2S \rightarrow HS^- + H^+$ is 10^7 less likely than the following reaction: $H_2SO_4 \rightarrow HSO_4^- + H^+$

ii. The reaction is



10^6 more likely than the reaction:



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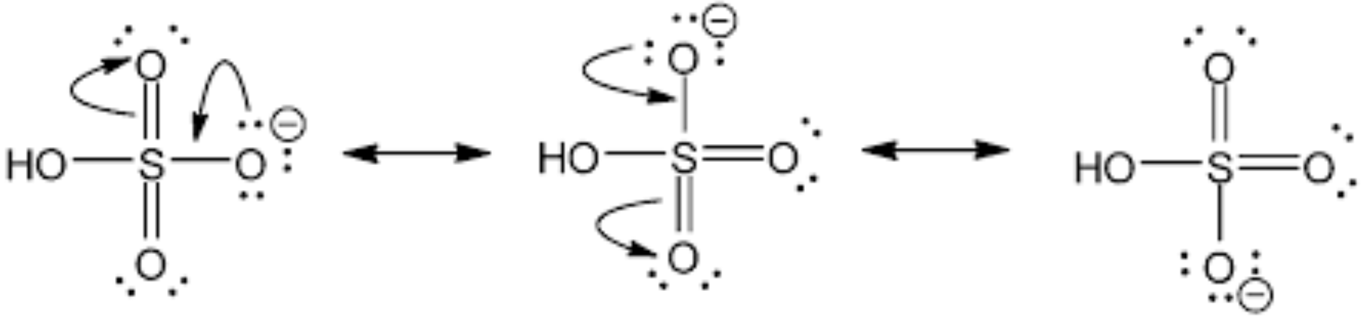
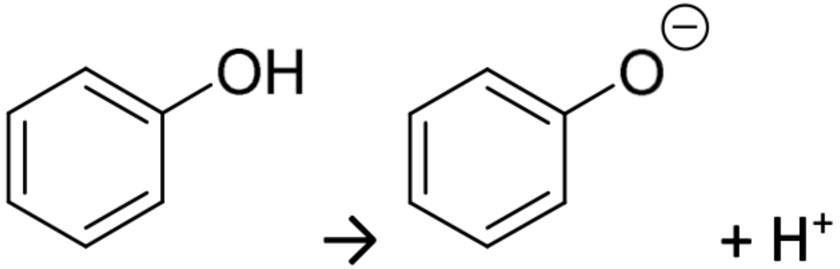
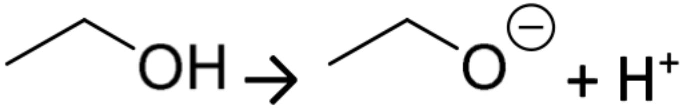
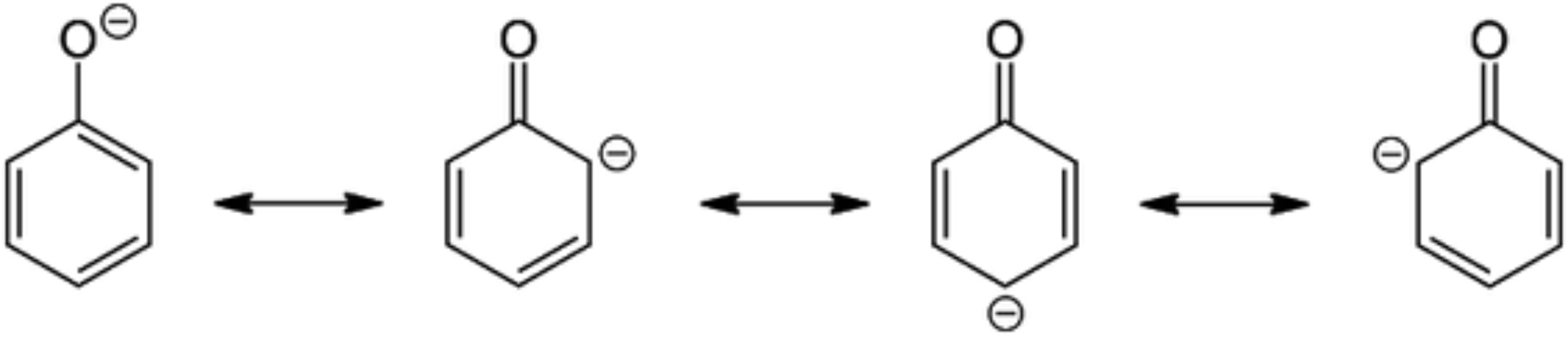


Electrostatic charges, whether positive or negative, are more stable when they are 'spread out' over a larger area: Resonance. This increases the stability of the conjugate base. In general, the greater the stability of the conjugate base, the stronger the acid.

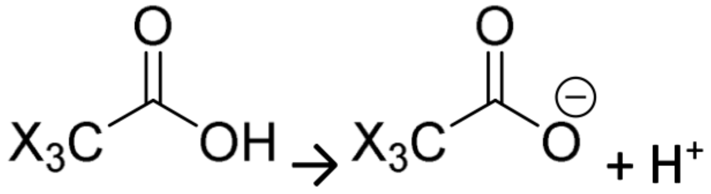
4) Explain the differences below using resonant forms:

ii. The reaction is  10^6 more

likely than the reaction: 



5) i) Consider **Q4i):** resonance is not the only explanation of this difference. Explain what other factors are relevant.

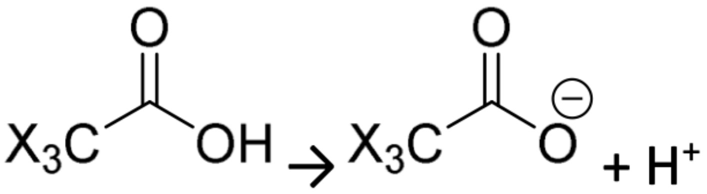
ii) The reaction  becomes increasingly likely in the following series: CX_3 : CH_3/CH_2Cl/ CH_2F/CCl_3/CF_3 (Likelihood follows the ratio: 1/74/149/12571/17143). Explain this trend.

iii) What role does resonance play in the reactions shown in 5ii)?

5) i) Consider **Q4i):** resonance is not the only explanation of this difference. Explain what other factors are relevant.

- **Bond Polarity:** In general, as the H—A bond polarity increases, to draw more electron density from H, the stronger the acid.

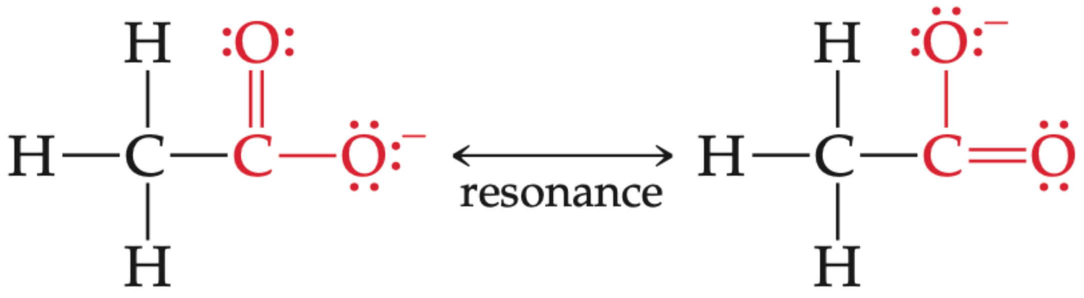
- Many oxyacids contain additional oxygen atoms bonded to the central atom Y. These atoms pull electron density from the O¬H bond, further increasing its polar- ity. Increasing the number of oxygen atoms also helps stabilize the conjugate base by increasing its ability to "spread out" its negative charge. Thus, the strength of an acid increases as additional electronegative atoms bond to the central atom Y.

5) ii) The reaction  becomes increasingly likely in the following series: CX_3 : CH_3/CH_2Cl/ CH_2F/CCl_3/CF_3 (Likelihood follows the ratio: 1/74/149/12571/17143). Explain this trend.

- Chlorine being a strong electron withdrawing group, pulls the negative charge towards itself by inductive effect so that the negative charge density on the oxygen atom is reduced, hence stabilizing the conjugate base of chloroacetic acid.

- Fluorine has a larger electronegativity, so it is able to stabilize the conjugate base better than chlorine

5) iii) What role does resonance play in the reactions shown in 5ii)?



Resonance increases the stability of the conjugate base n general, the greater the stability of the conjugate base, the stronger the acid.